escaping steam (ice condensers). Such containments therefore contain igniters in order to cause a controlled hydrogen burning.

In case of inerted PS-type containments, a large hydrogen production could still interfere with the blow-down process causing pressures exceeding the design limit. To prevent breaking the containment in case of overpressure all types of containments in many countries now connect by rupture disks or valves to *filtered venting systems* designed to capture most of the vented radioactive material.

In case of any event that clearly deviates form normal behavior of a reactor (ranging from observations with safety implications to catastrophes), the event should be reported to IAEA and the appropriate organizations in other countries. For reports to the press and to the general public of safety related events in nuclear power plants IAEA and OECD/NEA recommend the INES scale, see Table 19.6.

19.16. RADIOACTIVE REACTOR WASTE

In the "ideal" nuclear reactor all fission products and actinides produced are contained in the fuel elements. In all practical reactors there are four processes through which radioactivity leaves the reactor vessel; in all cases the carrier of activity is the coolant:

(i) induced radioactivity in the cooling medium (and moderator if separate);
(ii) corrosion products containing induced activities from construction materials;
(iii) leaked fission products and actinides from faulty fuel elements;
(iv) fission products of actinides deposited on surfaces in the core ("tramp uranium").

In all these respects pressurized and boiling water reactors present some problems of concern.

Although stringent efforts are made to contain all radioactivity formed by the nuclear reactions, it is unavoidable that small releases occur to the environment. Such releases, gaseous as well as liquid, are carefully monitored by measuring stations at the plant and at some locations around the plant. The authorities (nuclear regulatory commission, nuclear power inspectorate, radiation health board, etc.) set limits for such releases. Table 19.7 gives release values for a number of different reactor types and times, as reported by the United Nations. It should be noted that these are actual releases at that time. Usually the measured releases are a small fraction of the allowed limits and tend to decrease with time as more stringent limits are imposed by the regulatory bodies, improved equipment installed or better procedures adopted.

It is important to realize that the legally *permitted release* in fact sets the normal level of releases. Very much lower releases than permitted usually means increased operating costs. Hence, economic and safety considerations sets the practical release level to some chosen fraction of the legal limit, e.g. 1% or 0.1%. A 1% practical limit would give an ample margin to the legal limit, thereby reducing the risk that accidentally increased releases exceed the legal limit (which might have dire economic consequences for the plant operator).

Table 19.7 Average atmospheric and aquatic releases of radionuclides from various reactor types 1990–1997

Release (kBq/kWh)	PWR	BWR	GCR	HWR	RBMK	FBR
Airborne						
Noble gases	2.3	3.0×10^{1}	1.6×10^{2}	1.3×10^{2}	1.2×10^{2}	3.4×10^{1}
^{3}H	2.7×10^{-1}	1.0×10^{-1}	4.9×10^{-1}	5.6×10^{1}	3.0	5.6
^{14}C	1.9×10^{-2}	5.5×10^{-2}	1.1×10^{-1}	3.7×10^{-1}	1.5×10^{-1}	1.4×10^{-2}
^{131}I	2.9×10^{-5}	6.3×10^{-5}	1.0×10^{-4}	2.9×10^{-5}	8.0×10^{-4}	2.9×10^{-5}
Particulates	1.7×10^{-5}	3.0×10^{-2}	2.9×10^{-5}	5.7×10^{-6}	1.3×10^{-3}	7.4×10^{-4}
Liquid						
^{3}H	2.3	1.0×10^{-1}	2.9×10^{1}	4.7×10^{1}	1.3	2.0×10^{-1}
Other	1.5×10^{-3}	3.1×10^{-3}	6.9×10^{-2}	9.9×10^{-3}	6.3×10^{-4}	4.1×10^{-3}

(Data from UNSCEAR 2000)

Legal release limits are usually set so that a critical group of the general public should at most get an insignificant additional dose, typically 0.1 mSv/y. It is also customary to put a legal limit on the collective dose (usually per GW installed generating capacity) arising from operation of a nuclear power plant.

19.17. NUCLEAR FUSION

For more than 50 years scientists have studied methods of copying the fusion energy processes occurring in the sun and other stars in which hydrogen is transformed into helium. The availability of hydrogen and deuterium in the sea is so large that it will outlast any other non-renewable energy sources for any population growth rates: 1 liter of sea water contains deuterium with an energy content equivalent to 300 liter of gasoline (reaction 12.14).

At the first Geneva conference on the peaceful uses of atomic energy in 1955 there was strong sentiment that controlled thermonuclear reactors (CTR) would be in operation within 15 years. The optimism has gradually vanished as research has uncovered numerous technical difficulties. At present, it seems quite unlikely that such reactors will be in operation before the year 2050, although extensive research is being conducted towards CTR's, and great progress is being made. The following sections briefly describe the principal concepts, and some of the problems of controlled thermonuclear (fusion) power generation.

19.17.1. The controlled thermonuclear reactor

Fusion reactions are most easily achieved with hydrogen atoms because of the low coulomb barrier and favorable wave mechanical transmission factor. The threshold